

Errata and minor updates

Physical Geodesy: Global Elastic Deformation of the Earth

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1. Page 122, eq (2.489):
Change “ H_m ” to “ H_n ” after the second sum sign.

2. Page 123, eq (2.490), last line:
Change

$$\frac{1}{r \sin \theta} \frac{\partial B_\lambda}{\partial \lambda} + \frac{1}{r} A_\lambda B_\lambda$$

to

$$\frac{1}{r \sin \theta} A_\lambda \frac{\partial B_\lambda}{\partial \lambda} + \frac{1}{r} A_\lambda B_r$$

3. Page 123, eq (2.491), second line, last term in the parentheses:
Change “ B_λ ” to “ B_r ”
4. Page 154, below eq (3.118):
Change “ R ” to “ $\vec{R}$ ”
5. Page 236, from eq (4.40) to eq (4.43):

$$\begin{aligned} \Pi &> \left[1 - \frac{\alpha}{(N+1)N} \right] \cdots \left[1 - \frac{\alpha}{(i-3)(i-4)} \right] \left[1 - \frac{\alpha}{(i-1)(i-2)} \right] \\ &\quad \times \left[1 - \frac{\alpha}{(i+1)i} \right] \left[1 - \frac{\alpha}{(i+3)(i+2)} \right] \cdots \left[1 - \frac{\alpha}{(i+l+1)(i+l)} \right] \cdots \end{aligned} \quad (4.40)$$

Take the logarithm on both sides,

$$\begin{aligned} \ln \Pi &> \ln \left[1 - \frac{\alpha}{(N+1)N} \right] + \cdots + \ln \left[1 - \frac{\alpha}{(i-3)(i-4)} \right] \\ &\quad + \ln \left[1 - \frac{\alpha}{(i-1)(i-2)} \right] + \ln \left[1 - \frac{\alpha}{(i+1)i} \right] \\ &\quad + \ln \left[1 - \frac{\alpha}{(i+3)(i+2)} \right] + \cdots + \ln \left[1 - \frac{\alpha}{(i+l+1)(i+l)} \right] + \cdots \end{aligned} \quad (4.41)$$

As $\lim_{l \rightarrow \infty} \{\alpha / [(i+l+1)(i+l)]\} = 0$, we have

$$\lim_{i \rightarrow \infty} \left\{ \ln \left[1 - \frac{\alpha}{(i+l+1)(i+l)} \right] \middle/ \left[-\frac{\alpha}{(i+l+1)(i+l)} \right] \right\} = 1, \quad (4.42)$$

which can be considered as an alternative form of the formula $\lim_{x \rightarrow 0} [\ln(1+x)]/x = 1$. Hence, the infinite series at the right-hand side of (4.41) converges or diverges in the same way as the series

$$\begin{aligned} \frac{\alpha}{(N+1)N} + \cdots + \frac{\alpha}{(i-3)(i-4)} + \frac{\alpha}{(i-1)(i-2)} + \frac{\alpha}{(i+1)i} \\ + \frac{\alpha}{(i+3)(i+2)} + \cdots + \frac{\alpha}{(i+l+1)(i+l)} + \cdots, \end{aligned} \quad (4.43)$$

6. Page 280, left hand side of the second line of eqn (4.296):

Replace “ a_k ” by “ b_k ”

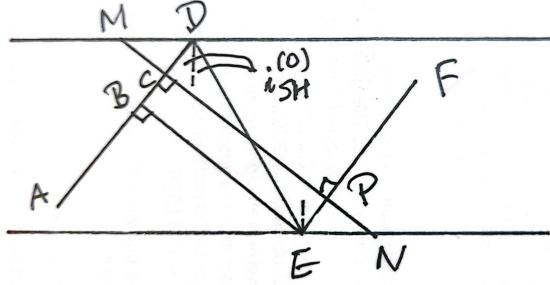
7. Page 281, in eqn (4.302):

Replace “ 2π ” by “ 4π ”

8. Page 283, line 1 below (4.314):

Change to “This formula holds for arbitrarily large N . Therefore, the positive term series” and change “ $\sum_{n=0}^N$ ” to “ $\sum_{n=0}^{\infty}$ ” in (4.315)

9. Page 514, replace Fig. 6.17 by (for making sure to avoid a probable copyright issue)



10. Page 642, line 2 from bottom. Correct to

Farrell (1972) and Guo et al (2001, 2004) for Sect. 7.8.

11. Page 753, eqn (9.59) should be

$$\delta s(\text{GR}) = \frac{R\bar{\rho}}{3} \sum_{n=2}^{n_{\max}^{\text{G2}}} \sum_{k=-n}^n \frac{2n+1}{1+k'_n} [\delta\Phi_n^k(\text{GR}) - \delta\Phi_n^k(\text{GA})] \bar{\mathcal{H}}_n^k(\theta, \lambda),$$